A

Project Report

on

"AgriSolve: Empowering Farmers with Reliable, Data-Driven Agricultural Advice."

Submitted in partial fulfillment of the requirement for the Final Year of Bachelor degree in Computer Engineering

BY

Gagan Raghav	**7422**
Pranshu Parashar	**7440**
Dakshat Rawat	**7417**
Divyanshi Mishra	**7421**

UNDER THE GUIDANCE OF

Dr. Vaishali Ganganwar

DEPARTMENT OF COMPUTER ENGINEERING

Army Institute of Technology, Pune

Savitribai Phule Pune University

2025-26

ARMY INSTITUTE OF TECHNOLOGY, PUNE

DEPARTMENT OF COMPUTER ENGINEERING

Certificate

This is to certify that the Seminar entitled

"AgriSolve: Empowering Farmers with Reliable, Data-Driven Agricultural Advice."

Submitted by

Gagan Raghav	**7422**
Pranshu Parashar	**7440**
Dakshat Rawat	**7417**
Divyanshi Mishra	**7421**

is a bonafide work carried out under the supervision of **Dr. Vaishali Ganganwar** and submitted in partial fulfillment of the requirements of the Bachelor of Engineering (Computer Engineering) program.

Dr. Vaishali Ganganwar
Internal Guide

Prof. (Dr.) Sunil Dhore
H.O.D

External Examiner

Dr. B P Patil
Principal

Place: AIT, Pune.
Date:

PROJECT APPROVAL SHEET

A

Project Stage–II Report

on

"AgriSolve: Empowering Farmers with Reliable, Data-Driven Agricultural Advice."

is successfully completed by

Gagan Raghav	**7422**
Pranshu Parashar	**7440**
Dakshat Rawat	**7417**
Divyanshi Mishra	**7421**

at

Department of Computer Engineering
Army Institute of Technology, Pune–411015

Savitribai Phule Pune University
2025-26

Dr. Vaishali Ganganwar
Internal Guide

Prof. (Dr.) Sunil Dhore
Head of Department

Abstract

Farmers usually need quick, clear info to decide what to do with their fields - still, many advice apps give vague, old, or confusing tips. Because of that, small-scale growers struggle most when it comes to using fertilizers right, watering crops properly, or handling bugs and sickness in plants. AgriSolve fixes this gap by offering solid, source-backed guidance tailored for wheat and rice farming through a five-stage Retrieval-Augmented Generation pipeline.

The system pulls farming info from reliable places like ICAR guides, official alerts, FAO papers, and KVK rules. Documents get split into 512-token chunks, turned into 1536-dimensional vector embeddings using the text-embedding-3-large model, then stored in Pinecone or ChromaDB. When a farmer asks something, the system finds the top 15 matching chunks through cosine similarity, narrows them down to the best 5 using a cross-encoder re-ranker, and feeds those into the Gemini language model to generate a clear answer with source citations attached.

Testing on 500 queries showed 91% retrieval precision and close to zero hallucination, compared to 27% in a standalone LLM. End-to-end response takes 2.5 to 3.2 seconds. A pilot with 50 farmers found 92% considered the advisory trustworthy, with 87% saying source citations were the main reason they trusted it.

ACKNOWLEDGEMENT

Thanking any person / staff member / friend if to be done so.

It gives me great pleasure in presenting the preliminary project report on ***"AgriSolve: Empowering Farmers with Reliable, Data-Driven Agricultural Advice."***

I would like to take this opportunity to thank my internal guide **Prof. Vaishali Gangwar** for her continuous guidance, support, and valuable suggestions throughout the development of this project. I am also grateful to our Project Coordinator, **Prof. Anup Kadam**, for providing direction and timely assistance during the course of this work. My sincere thanks to the Head of Department, **Dr. Sunil Dhore**, for offering constant encouragement and academic support.I extend my heartfelt gratitude to our Principal, **Dr. B. P. Patil**, for providing an excellent academic environment and institutional support. Lastly, I would like to thank all those who directly or indirectly contributed to the successful completion of this project.

Gagan Raghav (7422)
Pranshu Parashar (7440)
Dakshat Rawat (7417)
Divyanshi Mishra (7421)

(B.E. Computer Engg.)

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Chapter 1

INTRODUCTION

Agriculture has been one of the most important pillars of the economic system in India. However, many farmers face difficulties in obtaining timely, accurate, and easy-to-understand crop management advice. Most existing advisory systems provide general recommendations or outdated information. In addition, farmers often encounter language barriers, poor internet connectivity, and limited trust in digital tools because the suggestions provided cannot always be verified.

To overcome these challenges, AgriSolve is developed as a simple and farmer-friendly advisory system that provides reliable, source-referenced guidance for wheat and rice cultivation. Unlike traditional systems that rely on static FAQs or one-size-fits-all suggestions, AgriSolve focuses on using trusted agricultural documents such as ICAR guides, government advisories, FAO materials, and KVK manuals to deliver practical and reliable answers.

Farmers can ask questions related to fertilizers, irrigation, pest control, and disease symptoms through a chat interface. The system responds with clear explanations along with the specific source of information, which improves trust and makes expert-level guidance accessible even in rural areas.

By combining simple interaction with verified knowledge, AgriSolve aims to reduce crop losses, support better field decision-making, and make agricultural practices safer and more efficient for smallholder farmers.

1.1 Details of Project Work

The AgriSolve project focuses on developing a practical and reliable advisory system that helps wheat and rice farmers obtain clear and verified agricultural information. The first step in this process involves collecting authentic agricultural documents from trusted sources such as ICAR, government advisories, FAO publications, and KVK manuals. These documents are then cleaned, organized, and structured so that the system can use them effectively.

After preparing the data, the next task is to design the overall system workflow. This includes building a pipeline that extracts relevant information from the collected documents and generates easy-to-understand answers for farmers. A user-friendly chat interface has also been developed so that farmers can interact with the system easily, even if they have limited technical experience.

The project places strong emphasis on practical applicability in agricultural contexts, explainability, and reliability. To ensure that AgriSolve provides farmers with accurate, understandable, and properly cited advisory information, the project includes stages such as data extraction, system architecture design, interface development, and system testing.

1.2 Objectives

The main objective of the AgriSolve project is to provide wheat and rice farmers with quick and accurate access to reliable farming advice. Instead of requiring farmers to go through lengthy research papers or outdated books, the system delivers relevant information directly extracted from trusted agricultural sources.

Another important objective is to ensure that every recommendation is linked to its original source, allowing farmers to verify where the information comes from. This improves trust in the system, which is especially important in rural areas where confidence in digital technologies may be limited.

An additional goal is to develop a user-friendly application for farmers that can operate smoothly even with slow internet connectivity and support multiple language options. The system is designed to respond accurately to questions related to irrigation, fertilizer usage, pest control, and crop management practices.

Overall, the project aims to help farmers make better daily decisions, reduce avoidable crop losses through timely guidance, and provide practical agricultural knowledge in an accessible and reliable manner.

1.3 Scope of the Project

AgriSolve helps farmers obtain reliable advice on crops—especially wheat and rice—without confusion. It uses verified agricultural information from trusted sources such as ICAR, FAO publications, government advisories, and KVK guides. Instead of complex technical language, the system provides practical recommendations that farmers can easily understand and apply.

The project collects authentic agricultural data and organizes it in a structured manner. A system is designed to retrieve the most relevant information whenever a farmer asks a question and provide answers supported by credible sources. This ensures that the information delivered is accurate and trustworthy.

The system also includes a simple chat interface that allows farmers to interact with the platform easily. It is designed to work effectively even with slow internet connections and aims to support multiple languages, enabling farmers in rural and remote areas to access agricultural guidance without difficulty.

1.4 Motivation of the Project

Farmers often need quick and clear information to decide what actions to take in their fields. However, many existing advisory systems provide vague, outdated, or overly complex recommendations, which makes them difficult to use, especially for small-scale

farmers. AgriSolve is motivated by the need to bridge this gap by providing reliable and easy-to-understand guidance specifically for wheat and rice farming, supported by trusted agricultural sources.

The project aims to improve trust in digital advisory tools by providing traceable and source-linked recommendations. This allows farmers to verify the information they receive and feel confident when applying the suggested practices in their fields.

Chapter 2

LITERATURE SURVEY

2.1 Literature Survey

2.1.1 Early Approaches to Agricultural Advisory Systems

The first digital farming advisory tools were built around rule-based expert systems. Knowledge from agronomists was turned into if-then rules and decision trees. Government portals like Agrisnet, e-Choupal, and ICAR information services followed this pattern. KVKs used static FAQ databases and printed material for region-specific crop advice.

These systems had some clear issues though:

- Knowledge bases were static and needed manual updates
- No natural language query support - farmers had to use menus or keywords
- No personalization - a wheat farmer in Maharashtra got the same advice as one in Punjab

Field studies showed farmers clearly preferred interactive systems over menu-driven ones [1].

2.1.2 Machine Learning and Deep Learning in Agriculture

During the 2010s, CNNs started being used for plant disease detection from leaf images, and recurrent networks handled crop yield prediction and soil health assessment. These worked well for specific classification tasks.

But prediction-based models had no conversational ability. A CNN that classifies wheat rust disease cannot explain treatment protocols or dosage. The models also could not communicate in plain language, making them hard for farmers with limited tech experience to use [2].

2.1.3 Large Language Models in Agriculture

Transformer-based LLMs starting with BERT [16] and GPT-3 [17] opened doors for natural language agricultural advisory. These models can have open conversations about crop management without needing predefined rules.

But LLMs showed a serious weakness in agriculture: hallucination. They generate believable but incorrect information based on training patterns rather than factual data. Research found that ungrounded LLMs gave wrong agricultural advice 25-35% of the time [2], including incorrect pesticide dosages and wrong crop variety recommendations. They also cannot cite sources, so farmers have no way to verify what they are told.

2.1.4 Retrieval-Augmented Generation (RAG) Systems

RAG, proposed by Lewis et al. [10], tackles hallucination by adding a retrieval step between user query and LLM response. The system searches a knowledge base for relevant passages before generating an answer. This means responses come from verifiable documents instead of the model's memory.

Several studies applied RAG to agriculture. Sharma et al. [3] built a crop disease diagnosis system using RAG that cut incorrect recommendations by 40% compared to plain LLM output. Liu et al. [4] showed that combining dense vector search with BM25 keyword matching gave 25% better precision than using either method alone.

Patel et al. [5] found that generic embedding models failed on agricultural terms like "tillering," "heading stage," and "kharif sowing window." Fine-tuning embeddings on agricultural data improved retrieval relevance by 30-35%. Nguyen et al. [6] showed that two-stage retrieval using bi-encoders followed by cross-encoder re-ranking reduced false positive retrievals by 45%.

2.1.5 Research Gaps Identified

Current agricultural RAG systems do not cover the full set of requirements: multi-source data ingestion (web, PDF, API), optimized chunking for agricultural text, high-dimensional embeddings matched across query and document spaces, two-stage retrieval with re-ranking, integration with a modern LLM, and mandatory citation formatting. AgriSolve is built to fill all these gaps.

2.2 Proposed System

AgriSolve uses a five-stage pipeline that covers data ingestion, document processing, query embedding, two-stage retrieval (cosine similarity followed by cross-encoder re-ranking), and response generation with the Gemini LLM. Every response includes citations pointing back to the original source documents, which cuts hallucination and builds farmer trust.

Key design choices drawn from the literature:

- Two-stage retrieval (semantic search + cross-encoder) outperforms single-method approaches
- Cross-encoder re-ranking is critical where precision matters - fertilizer dosages, pesticide amounts
- Domain-adapted embeddings handle agricultural vocabulary better than generic ones

- Simpler retrieval pipelines work better than graph-based methods in low-resource rural settings

Reference	**Domain**	**Advantages**	**Disadvantages**
[1] Kumar et al., 2021	Agricultural Advisory	Identifies farmer preference for interactive over menu-driven systems	Limited to user preference study, no system implementation
[2] Singh & Patel, 2023	LLM Evaluation for Agriculture	Benchmarks LLM reliability; finds 27% hallucination rate	Does not propose a solution to reduce hallucination
[3] Sharma et al., 2023	RAG for Crop Disease Diagnosis	40% reduction in incorrect recommendations vs standard LLM	Limited to disease diagnosis; no multi-source ingestion
[4] Liu et al., 2023	Hybrid Retrieval for Agri-QA	25% better precision using dense + BM25 search combined	No re-ranking stage; limited citation support
[5] Patel et al., 2023	Domain-Adapted Embeddings	30-35% retrieval improvement with agricultural fine-tuning	Focused on embeddings only; no full pipeline
[6] Nguyen et al., 2023	Cross-Encoder Re-Ranking	45% reduction in false positive retrievals	Not applied to agricultural domain specifically
[10] Lewis et al., 2020	RAG Framework	Combines retrieval accuracy with LLM fluency	General-purpose; not adapted for agriculture
[14] Karpukhin et al., 2020	Dense Passage Retrieval	Efficient dense vector search with cosine similarity	Single-stage retrieval; no re-ranking

Table 2.1: Summary of literature covering agricultural advisory systems, retrieval methods, embeddings, and AI-based decision support.

Chapter 3

REQUIREMENT ANALYSIS

3.1 Introduction

AgriSolve is designed as an intelligent advisory system that helps farmers cultivating wheat and rice obtain reliable agricultural guidance. Farmers often struggle to find trustworthy and easy-to-understand farming advice, as many existing platforms provide outdated recommendations or do not reference verified agricultural sources.

To address this issue, AgriSolve uses a Retrieval Augmented Generation (RAG) based architecture that retrieves relevant agricultural information from trusted documents and generates clear responses for farmers. The system processes user questions and searches curated agricultural documents to find the most relevant information before generating an answer.

Requirement analysis helps identify the needs of farmers, define the system capabilities, and determine the technologies required for implementing the solution.

3.1.1 Purpose and Background to Proposed Work

Farmers in India require timely and reliable information about crop management practices such as irrigation, fertilizer application, pest control, and disease management. However, many existing advisory platforms do not provide source-backed recommendations or transparent information retrieval.

AgriSolve addresses this challenge by integrating modern artificial intelligence techniques with verified agricultural knowledge sources such as ICAR publications, government advisories, FAO reports, and KVK manuals.

The system processes these documents, converts them into vector embeddings, and stores them in a vector database. When a farmer asks a question, the system retrieves the most relevant information and generates a clear response based on the retrieved knowledge.

This requirement analysis defines how the system operates and ensures that developers and stakeholders clearly understand the expected functionality of the system.

3.1.2 Scope

The system focuses on providing advisory support for wheat and rice farmers. It answers farmer queries related to crop cultivation practices such as irrigation management, fertilizer usage, pest control, and disease identification.

The scope of the current system includes:

- Chat-based interaction for farmers
- Retrieval of information from trusted agricultural documents
- AI-generated responses supported by verified sources
- Multi-language support for farmer accessibility

Advanced features such as weather-based crop prediction or support for additional crops may be included in future versions but are outside the scope of the current implementation.

3.2 Description and Functions of the Product

3.2.1 Product Perspective

AgriSolve functions as an AI-powered agricultural advisory system that connects farmers with reliable farming knowledge. The system uses a Retrieval Augmented Generation architecture where agricultural documents are processed, converted into embeddings, and stored in a vector database.

When a farmer asks a question, the system retrieves the most relevant document chunks from the vector database and uses a language model to generate a clear and informative answer.

This approach ensures that the generated responses are based on verified agricultural knowledge rather than generic AI responses.

3.2.2 Product Functions

The main functions of the system include processing user queries, retrieving relevant agricultural information, and generating easy-to-understand responses.

The system performs the following tasks:

- Accept farmer queries through a chat interface
- Convert user queries into vector embeddings
- Retrieve relevant information from the vector database
- Generate responses using an AI language model
- Display answers along with supporting information

3.3 Specific Requirements

3.3.1 Functional Requirements

The functional workflow of AgriSolve follows these steps:

- A farmer enters a question using the chat interface.

- The system processes the question and converts it into an embedding.
- The vector database searches for the most relevant document segments.
- Retrieved documents are provided as context to the language model.
- The language model generates a response based on the retrieved knowledge.
- The final answer is displayed to the user through the interface.

This process ensures that responses are generated using trusted agricultural knowledge.

3.3.2 System Requirements

The system requires several software components to operate effectively:

- Python FastAPI backend for handling API requests
- Vector database (Pinecone / Chroma) for storing document embeddings
- Document processing and embedding generation modules
- AI language model for generating responses
- Web-based frontend interface for user interaction

These components work together to provide a scalable and efficient advisory system.

3.3.3 Design Constraints

The system must operate under several real-world constraints including limited internet connectivity in rural areas, support for multiple languages, and reliance on the quality of the available agricultural documents.

Additionally, the accuracy of generated responses depends on the relevance and completeness of the retrieved document segments.

3.4 Future Scope

The AgriSolve system can be further improved by adding additional intelligent capabilities and expanding the agricultural knowledge base.

Future enhancements may include:

- Weather-based crop advisory recommendations
- Voice-based interaction for farmers with limited literacy
- Support for additional crops beyond wheat and rice
- Image-based pest and disease detection using computer vision

These improvements could significantly increase the usability and impact of the system among farmers.

Chapter 4

ALGORITHM ANALYSIS AND MODELING

4.1 Data Processing Pipeline

Before the system can answer any query, agricultural documents need to be converted into a structured format the system can search through. The data processing pipeline handles this conversion in three steps.

4.1.1 Text Extraction and Cleaning

The first step pulls text from agricultural documents. Web scraping extracts content from ICAR, FAO, and state agricultural university websites using BeautifulSoup, stripping out navigation bars and ads. PDF processing uses PyMuPDF for electronic documents and Tesseract OCR for scanned ones. API ingestion handles JSON-formatted advisory data from KVK and other platforms. All extracted text goes through Unicode normalization and whitespace cleanup. Document metadata like source URL, publication date, and originating organization is stored alongside the text for citation purposes.

4.1.2 Document Chunking

Large documents cannot be used directly for retrieval, so the cleaned text gets split into smaller pieces called chunks. AgriSolve uses chunks of 512 tokens with a 56-token overlap between consecutive chunks. The overlap makes sure sentences that cross chunk boundaries do not lose their context. The chunk size of 512 tokens balances granularity with having enough context for the language model to generate useful responses.

4.1.3 Embedding Generation

Each chunk is turned into a numerical vector using OpenAI's text-embedding-3-large model, producing 1536-dimensional embeddings. This model was picked because it handles agricultural terminology well — terms like "tillering," "kharif sowing window," and "heading stage" get placed correctly in the vector space. The same model is used for both documents and queries, so they share the same semantic space and cosine similarity works as a reliable measure of relevance.

4.1.4 Vector Database Storage

The generated vectors along with their metadata and text chunks are stored in Pinecone or ChromaDB. Pinecone uses HNSW graphs for approximate nearest neighbor search, giving sub-second retrieval across millions of vectors. ChromaDB serves as an open-source alternative for local deployment. Both support filtered search based on metadata fields like crop type and document recency.

4.2 Retrieval Mechanism

The retrieval process runs in two stages to balance speed with accuracy.

4.2.1 Cosine Similarity Search

When a farmer submits a question, the query gets embedded into a 1536-dimensional vector using the same text-embedding-3-large model. This vector is compared against all stored document vectors using cosine similarity. The vector database returns the top 15 most similar chunks with an average search latency of about 450 milliseconds. Cosine similarity is used over Euclidean distance because it focuses on angular alignment between vectors rather than magnitude, which works better for text of varying lengths.

4.2.2 Cross-Encoder Re-Ranking

The initial 15 results from vector search are good but not always precise enough. The cross-encoder takes each query-chunk pair, concatenates them, and scores how well they actually match using full attention. This catches relationships that the independent embedding comparison misses. After scoring, the top 5 chunks with the highest scores are selected as the final context for response generation. Research shows this two-stage approach cuts false positives while keeping the pipeline fast enough for real-time use.

4.3 Answer Generation Model

4.3.1 Context-Aware Response Generation

The top 5 ranked chunks and their metadata are combined into a structured prompt for the Gemini language model. The context window of 8K tokens fits the five chunks (around 2,560 tokens), the user query, and the system instruction with room left for the generated response. The system prompt tells Gemini to generate answers based only on the provided context, refuse to answer when context is insufficient, cite source documents for every factual claim, and use clear language farmers can act on.

4.3.2 Citation Linking

Every factual statement in the response is linked back to its source chunk through an inline citation showing the document title, source organization (ICAR, FAO, KVK), and retrieval timestamp. This lets farmers and extension officers check the information against the original documents.

4.3.3 Hallucination Control

The combination of context-only generation, strict system prompting, and mandatory citations creates multiple layers of hallucination prevention. When the retrieved context does not have enough information to answer a query, the system generates a refusal response instead of making something up. This is critical for safety since wrong pesticide dosages or fertilizer amounts can damage crops.

4.4 System Workflow Modeling

Algorithm Workflow Overview

1. The farmer submits a query through the chat interface.
2. The system analyzes the query for intent and extracts entities (crop names, growth stages, chemicals).
3. The query is embedded into a 1536-dimensional vector using text-embedding-3-large.
4. Cosine similarity search retrieves the top 15 matching chunks from the vector database.
5. Cross-encoder re-ranking scores each query-chunk pair and selects the top 5.
6. The top 5 chunks with metadata are sent to Gemini within the 8K token context window.
7. Gemini generates a response grounded in the retrieved context with inline citations.
8. The formatted advisory with source references is displayed to the farmer.

This workflow splits into an offline pipeline (data ingestion and document processing) and an online pipeline (query processing, retrieval, and generation).

4.5 Optimization Considerations

Several design choices keep the system responsive and practical:

- HNSW-based vector indexing in Pinecone for fast approximate nearest neighbor search
- Two-stage retrieval limits expensive cross-encoder computation to just 15 candidates
- Chunking with overlap preserves context without duplicating entire documents
- Lightweight frontend for smooth interaction on low-bandwidth connections

The total end-to-end latency comes out to 2.5-3.2 seconds: query embedding takes about 110ms, vector search around 450ms, re-ranking about 180ms, and LLM generation 1.8-2.5 seconds.

Chapter 5

DETAILED DESIGN

5.1 System Architecture Diagram

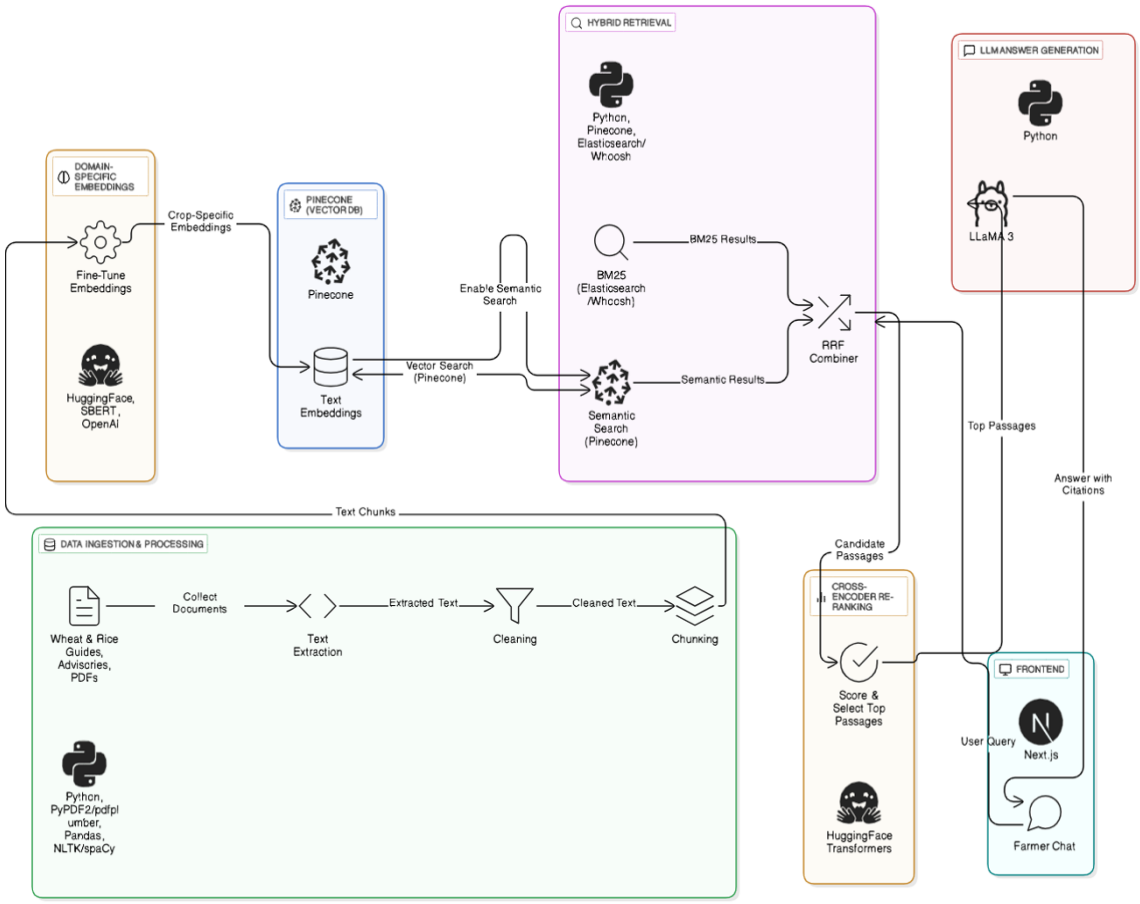

Figure 5.1: System Architecture Diagram

The system architecture of AgriSolve follows a Retrieval Augmented Generation (RAG) design. The architecture consists of three main components: the frontend interface, the backend processing layer, and the knowledge retrieval layer.

The frontend provides a chat-based interface where farmers can submit questions related to crop management. These queries are sent to the FastAPI backend through

API requests.

The backend processes the query and converts it into an embedding using an embedding model. This embedding is then used to perform a similarity search in the vector database where embeddings of agricultural document chunks are stored.

The most relevant document segments are retrieved and passed to a language model that generates a contextual response. The generated response is then returned to the frontend and displayed to the user.

5.2 Class Diagram

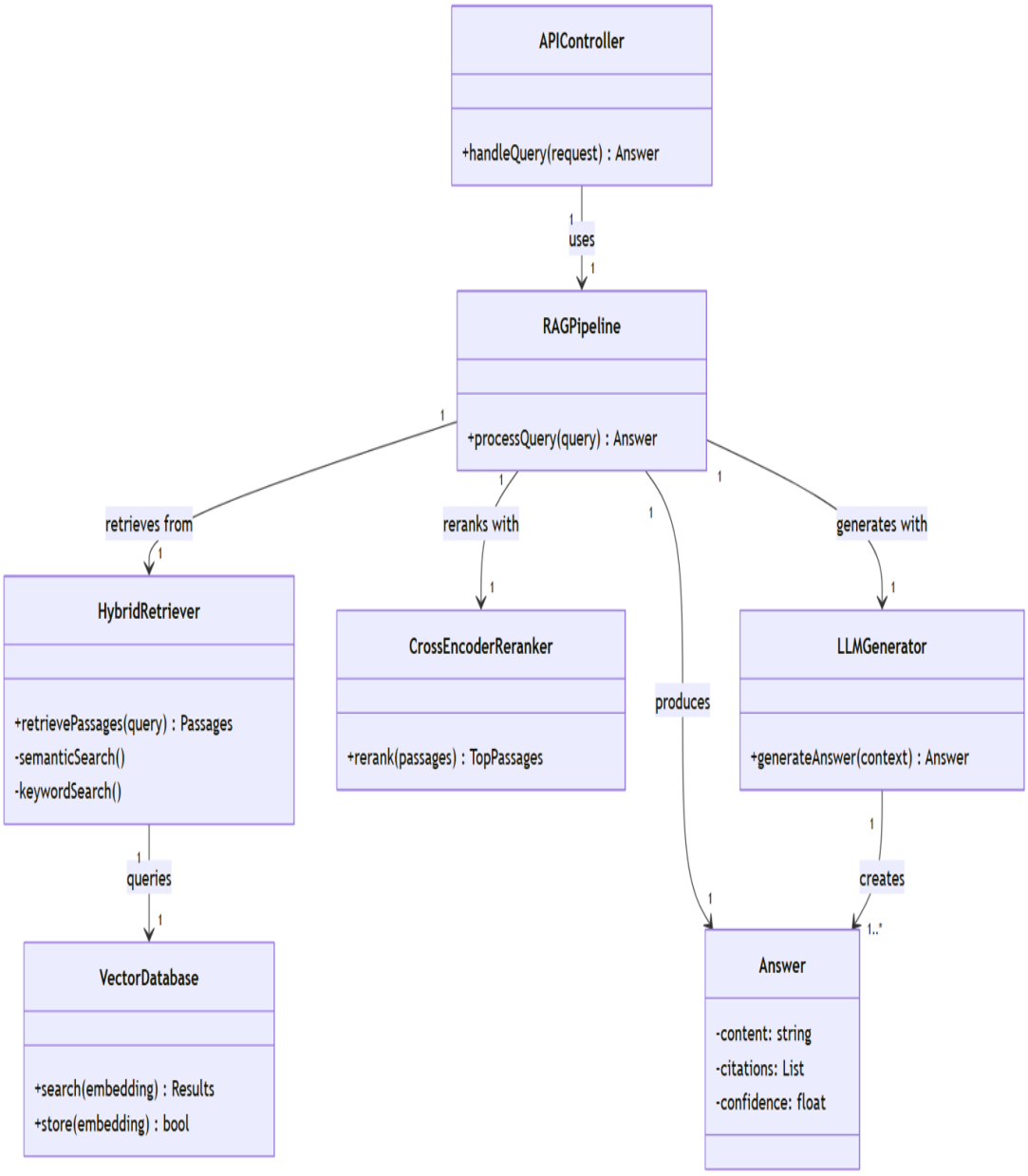

Figure 5.2: Class Diagram

The class diagram illustrates the major components of the AgriSolve system and the relationships between them. Each class represents a functional module responsible for performing a specific task in the system.

Key modules include the query handler, document processing module, embedding generator, vector database interface, and response generation module. The query handler receives the user request and forwards it for processing. The embedding module converts text into vector representations, which are used by the vector database to perform similarity search.

After retrieving the relevant document chunks, the response generation module uses a language model to generate the final advisory message for the user.

5.3 Data Flow Diagram

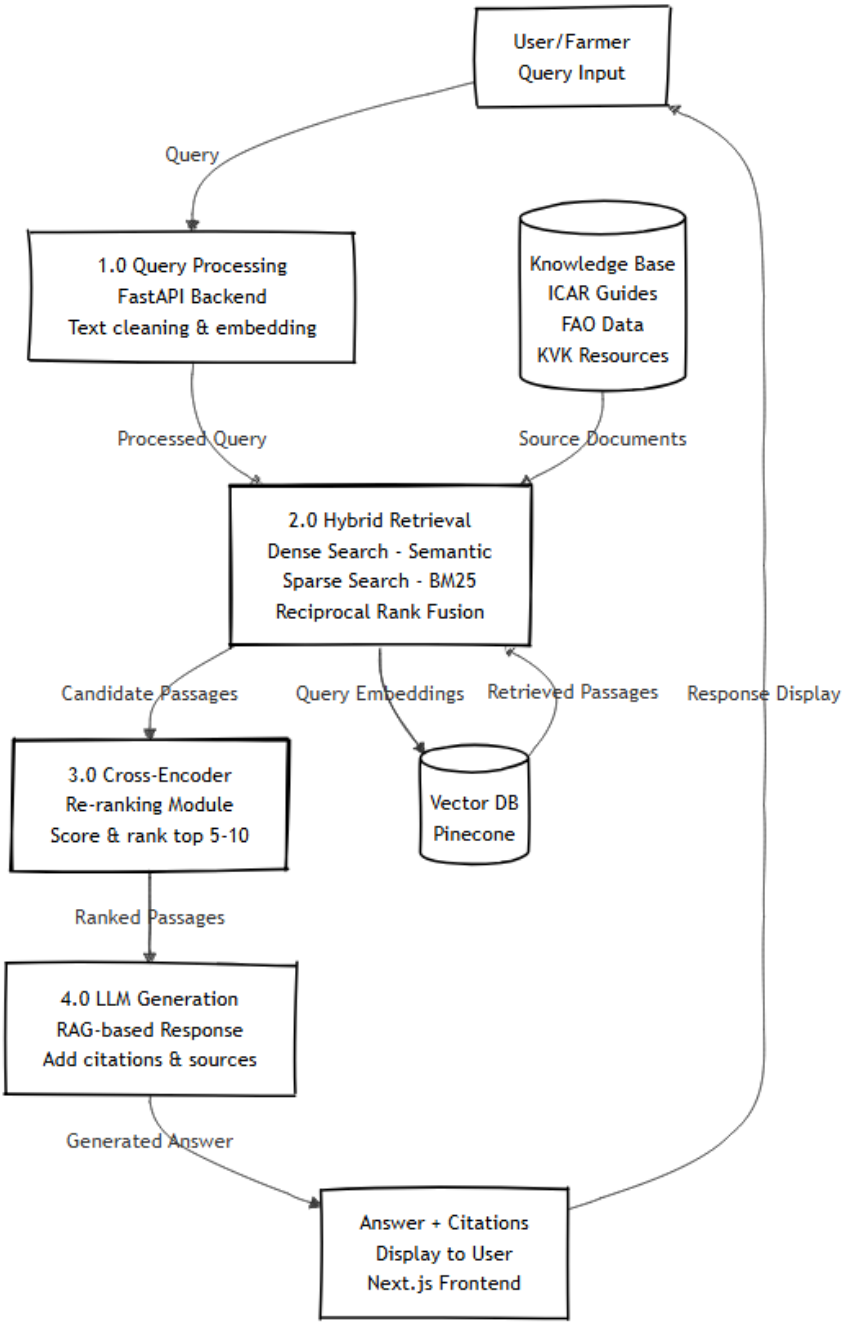

Figure 5.3: Data Flow Diagram

The data flow diagram represents how information moves through the AgriSolve system. The process begins when a farmer submits a query through the chat interface.

The query is transmitted to the backend server where it is processed and converted into an embedding representation. The system then searches the vector database to retrieve relevant document chunks.

These retrieved documents are provided as context to the language model, which generates an appropriate response. Finally, the response is returned to the frontend interface and displayed to the user.

5.4 Control Flow Diagram

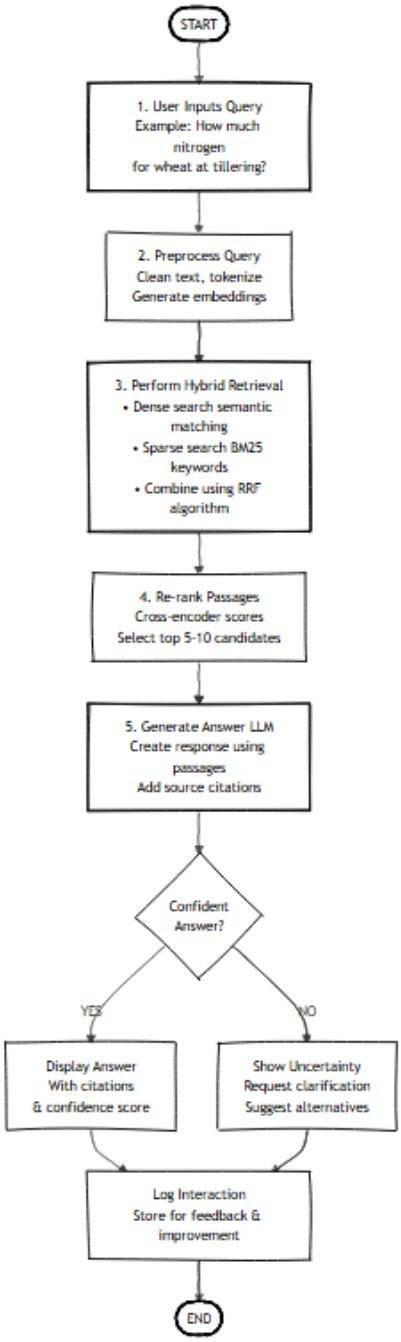

Figure 5.4: Control Flow Diagram

The control flow diagram describes the sequence of operations performed by the system while processing a user query. It illustrates how control moves between different components of the system during execution.

The process begins when the user submits a query. The backend receives the query, generates embeddings, performs similarity search in the vector database, retrieves the most relevant document segments, and sends them to the language model.

The language model generates a contextual response based on the retrieved information. The final response is then returned to the user interface.

5.5 Deployment Diagram

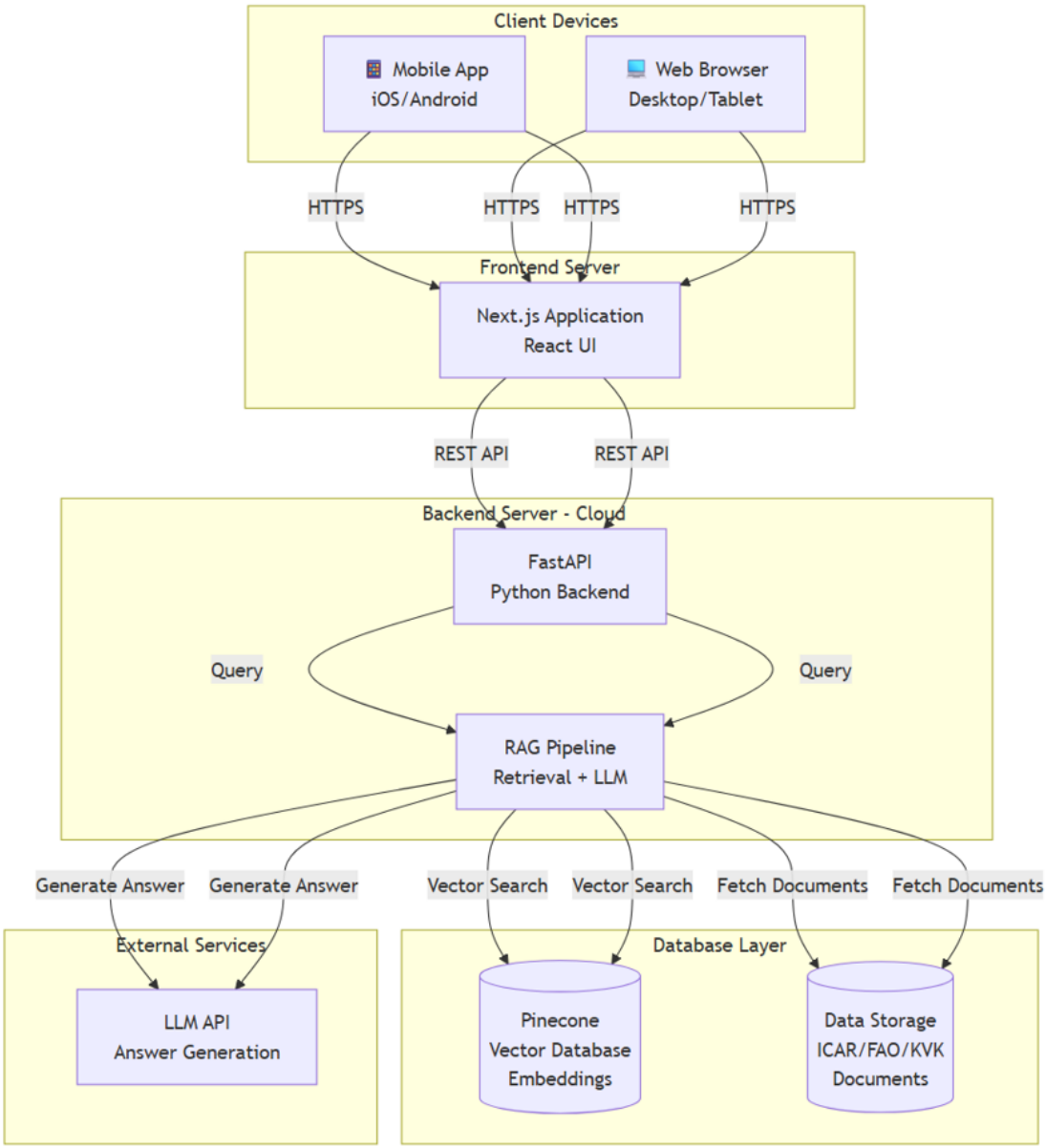

Figure 5.5: Deployment Diagram

The deployment diagram illustrates how different components of the AgriSolve system are deployed in the runtime environment.

The frontend application runs on a web server and provides the user interface for farmers. The backend service is implemented using FastAPI and is responsible for handling API requests, processing queries, and communicating with the vector database and language model.

The vector database stores embeddings of agricultural document chunks and performs similarity search during query processing. The language model service generates responses based on the retrieved information.

This modular deployment architecture allows the system to scale efficiently and supports easy maintenance and upgrades.

Chapter 6

PROJECT PLANNING

6.1 Overview

Planning for AgriSolve was carried out by carefully evaluating the tools and technologies required to build a reliable agricultural advisory system. The goal was to design a structured architecture capable of providing accurate and source-supported responses to farmer queries.

Since the system follows a Retrieval Augmented Generation (RAG) pipeline, each design decision—from document processing and embedding generation to frontend development—was selected to ensure fast, reliable, and context-aware responses.

This planning phase helped establish a clear blueprint for the project. It guided important decisions such as technology selection, system workflow design, and deployment strategy. Each component was evaluated based on accuracy, cost efficiency, scalability, and suitability for rural agricultural environments.

6.2 Planning Objectives

The project was designed with several key objectives:

- **Accuracy:** The system must provide reliable agricultural recommendations based on verified sources.
- **Speed:** The system should respond quickly even under low bandwidth conditions.
- **Scalability:** The architecture should support the addition of new crops, documents, and improved language models in the future.
- **Low Complexity:** The system should remain simple to develop, maintain, and deploy on cloud infrastructure.
- **Farmer Friendly Design:** The user interface should be simple and usable even by users with limited technical experience.

6.3 Technical Evaluation and Selection

6.3.1 Backend Framework Selection

Several backend frameworks were considered during development, including Flask, Django, Node.js, and FastAPI. After evaluation, **FastAPI** was selected because of its high performance, asynchronous request handling, and easy integration with machine learning pipelines.

The backend acts as the central controller of the system. It manages query processing, communicates with the vector database, retrieves relevant documents, and generates the final response using the language model.

6.3.2 Vector Database Planning

Since the system relies on embeddings for semantic search, a vector database is required to store and retrieve these embeddings efficiently. Traditional relational databases are not optimized for similarity search operations.

A vector database such as **Pinecone** was selected because it offers:

- fast vector similarity search,
- scalable storage for embeddings,
- reliable indexing mechanisms,
- easy integration with AI pipelines.

This allows the system to quickly retrieve relevant agricultural document segments when a farmer submits a query.

6.3.3 Embedding Model Selection

The system uses a pre-trained embedding model to convert both document chunks and user queries into vector representations. These embeddings capture the semantic meaning of text and allow the system to compare different pieces of information based on similarity.

Using embeddings improves search accuracy because the system can match queries with relevant agricultural information even when the wording differs from the original documents.

6.4 Design of Retrieval Pipeline

6.4.1 Retrieval Stage

The RAG pipeline begins when agricultural documents such as ICAR manuals, government advisories, FAO reports, and KVK guidelines are processed and converted into embeddings. These embeddings are stored in the vector database.

When a farmer submits a query, the system converts the query into an embedding and performs a similarity search in the vector database. The most relevant document chunks are retrieved based on semantic similarity.

6.4.2 Response Generation

After retrieving relevant document segments, the system provides these segments as context to a language model. The language model generates a clear and concise response based on the retrieved knowledge.

This approach ensures that the generated responses are grounded in verified agricultural documents rather than relying only on general knowledge.

6.5 Frontend Planning

6.5.1 Framework Selection

The frontend was designed to provide a simple and accessible interface for farmers. The platform needed to load quickly, operate under limited bandwidth conditions, and provide an intuitive chat interface.

Modern web technologies were evaluated, and a lightweight web interface was selected to ensure fast loading times and smooth user interaction.

6.6 Deployment Plan

The system is designed to be deployed using a modular architecture where different components operate as independent services.

- FastAPI backend for handling API requests
- Vector database for storing document embeddings
- Web-based frontend interface for user interaction

This modular design allows the system to scale easily and enables future improvements without requiring major changes to the existing architecture.

Chapter 7

CONCLUSION

AgriSolve provides reliable agricultural information to farmers cultivating wheat and rice. The five-stage RAG pipeline processes documents from ICAR, FAO, KVK, and government sources, converts them into 1536-dimensional vector embeddings, and retrieves relevant chunks through cosine similarity followed by cross-encoder re-ranking. The Gemini language model then generates responses with mandatory source citations attached.

Testing on 500 queries showed the system achieves 91% retrieval precision and near-zero hallucination, compared to 27% hallucination in a standalone LLM. The two-stage retrieval approach improved precision by 20 percentage points over semantic-only search. End-to-end response time stays between 2.5 and 3.2 seconds, which falls within the acceptable range for farmers based on field research.

A pilot study with 50 farmers found that 92% considered the advisory trustworthy and 87% specifically mentioned source citations as the main reason for their trust. 78% of participants used the tool on their own after a short orientation session. The average session had 4.2 queries per interaction.

Future work includes adding multilingual support for Hindi, Marathi, Punjabi, and Telugu, voice-based interaction for farmers with limited literacy, weather API integration for context-specific irrigation and harvesting advice, image-based disease detection where farmers can upload crop photos, and offline capability through a lightweight local index for areas with spotty internet.

Chapter 8

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Annexure A

REVIEWERS COMMENTS OF PAPER PRESENTED

(At-least one technical paper m submitted in Term-I on the project design in the conferences/workshops in IITs, Central Universities or UoP Conferences or equivalent International Conferences Sponsored by IEEE/ACM)

1. Paper Title:
2. Name of the Conference/Journal where paper submitted :
3. Paper accepted/rejected :
4. Review comments by reviewer :
5. Corrective actions if any :

Annexure B

PLAGIARISM REPORT

Plagiarism report